ECON 0100 | Part D

Vignette D1

Q1 (1 of 3) | Shrake Fishery

The Shrake is a powerfully magical fish with significant interest as a potent ingredient in potions. Since their spines easily damage fishing nets, only two massive cooporate firms (confidentially called Firm A and Firm B) remain in the market after independently developing highly secretive technology for harvesting Shrake. Both firms have the ability to harvest vast quantities of Shrake but can make substantial profit even with low harvests. At low capacities (L) the fishery could support both firms, but at high capacities (H) overfishing would lead to population collapse in future years. Following these two fishing strategies, the lifetime profits (in Galleons) for the are summarized by the following payoff matrix.

		Firm B: L	Firm B: H
Firm A	L	10, 6	2, 10
	H	18, 1	3, 3

(1)

Recall Best Response: The best Strategy for one firm, given what we think the other firm may do.

A. What is Firm A's best response?

$$BR_A = \begin{cases} H & \text{IF, } L \\ H & \text{IF, } H \end{cases}$$

Why? Because if firm B picks L, firm A will pick H, because the payoff would be 18 galleons as opposed to 10 if they also picked L. The same logic applies if B picks H, (3>2).

B. What is Firm B's best response?

$$BR_B = \begin{cases} H & \text{IF, } L \\ H & \text{IF, } H \end{cases}$$

Why? Because if firm A picks L, firm B will pick H because the payoff would be 10 galleons as opposed to 6 if they also picked L. The same logic applies if A picks H, (3>1).

C. Find all Nash Equilibium.

Nash = (H, H)

Why? Because at (H, H), neither firm has an incentive to change their Strategy, given what the other firm is doing.

D. Find the socially efficient strategies.

SE = (H, L)

Why? The socially efficient strategy is the one that maximizes the welfare or benefits combined for both firms, in this case it is (H, L) because 19 > 16, 12, 6.

Q2 (2 of 3) | Shrake Fishery

Fearing a total collapse of the Shrake fishery, the Ministry of Magic hired Remus Lupin to develop a policy solution to the looming issue. Remus offered a few proposals but the policy chosen by the Ministry was to distribute a fishing permit to both Firm A and Firm B to harvest at low capacity (L) each season. This would lead to a sustainable population of Shrake into future seasons. Since there are only two firms supplying Shrake, the market is highly visible to regulators making the permits very enforceable. Any firm in violation, choosing a high capacity (H), would be required to pay a fine equal to half their seasonal profit. The following matrix summarizes the two firms' lifetime profits after the enforceable permits.

		Firm B: L	Firm B: H
Firm A	L	10, 6	2, 5
	H	9, 0.5	1.5, 1.5

(2)

A. What is Firm A's best response? Why? If firm B picks L, firm A will also choose L because the payoff is higher (10 > 9). The same logic applies if B picks H, A will choose L again because (2 > 1.5).

$$BR_A = \begin{cases} L & \text{IF, } L \\ L & \text{IF, } H \end{cases}$$

B. What is Firm B's best response? Why? If firm A picks L, firm B will also choose L because the payoff is higher (6 > 5). The same logic applies if A picks H, B will also choose H because (1.5 > 0.5).

$$BR_B = \begin{cases} L & \text{IF, } L \\ H & \text{IF, } H \end{cases}$$

C. Find all Nash Equilibium. Why? At the Strategy (L, L), neither firm has an incentive to change their own strategy, given what the other is doing.

Nash = (L, L)

Q3 (of 3) | Study Room Dynamics

Students in Ravenclaw and Hufflepuff share a secondary study room, which is open to all students from both houses at all hours. Every year the heads of house from both houses organize the sharing of cleaning duties. Cleaning improves the quality of the study room for members of both houses, but the chore is inconvenient. If both houses clean, the study room is a fantastic resource. But cleaning takes precious time away from studying, so neither house prefers doing it. The following matrix represents the benefits of each possible strategy combination.

		Hufflepuff: Clean	Hufflepuff: Don't Clean
Ravenclaw	Clean	10, 10	6, 12
	Don't Clean	12, 6	8, 8

(3)

A. What is Ravenclaw's best response? Why? If H chooses C, R's best response is to choose DC, because a payoff of 12 is better than 10. The same logic applies if H chooses DC, R will also choose DC because (8>6).

$$BR_R = \begin{cases} DC & \text{IF, } C \\ DC & \text{IF, } DC \end{cases}$$

B. What is Hufflepuff's best response? Why? If R chooses C, H's best response is to choose DC, because a payoff of 12 is better than 10. The same logic applies if R chooses DC, H will also choose DC because (8,6).

$$BR_H = \begin{cases} DC & \text{IF, } C \\ DC & \text{IF, } DC \end{cases}$$

C. Find all Nash Equilibium. Why? At the strategy (DC, DC), neither firm has an incentive to change their own strategy based off of what the other is doing because the payoff would make them worse off.

Nash = (DC, DC)

D. Find the socially efficient strategies. The socially efficient strategy yields the maximum total benefit or payoff for both firms combined. 20 > 18, 16.

SE = (C, C)